Project 5: E-Commerce Return Risk Prediction & Operational Dashboard

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Abstract

This report details the end-to-end development of an analytical and predictive machine learning pipeline aimed at mitigating revenue loss from product returns in the e-commerce sector. Utilizing a dataset containing one million Amazon transaction records, the project bridges the gap between backend data science and frontend business intelligence. A Logistic Regression classification model was trained to calculate the precise probability of return for individual orders based on pricing, discounts, and categorical features. The predictive output was subsequently integrated into a Power BI enterprise dashboard, empowering operational managers with real-time tracking of categorical return rates and a prioritized ledger of high-risk transactions.

Tools & Technologies Used

- Python 3.x
- Google Colab
- Pandas & NumPy
- Scikit-Learn (Logistic Regression)
- Power BI Desktop
- DAX (Data Analysis Expressions)
- Power Query

Implementation Steps

1. Data Preprocessing & Exploratory Aggregation

The initial phase was conducted in Google Colab using Python's `pandas` library. The one-million-row Amazon dataset was ingested, and the target variable (`is_returned`) was standardized into a binary format (1 for returned, 0 for retained). Initial data aggregation replicated SQL-style grouping to calculate aggregate return rates across top-level product categories, providing a baseline understanding of departmental performance.

2. Feature Engineering & Machine Learning

Categorical variables such as product category were transformed using one-hot encoding. Continuous variables, including `price`, `discount`, `final_price`, `rating`, and `shipping_time_days`, were normalized using a `StandardScaler` to ensure uniform model weighting. A Logistic Regression model was initialized and fitted to the scaled features. Using the `predict_proba` function, the model generated a specific "Return Risk Score" (ranging from 0.00 to 1.00) for every individual order, predicting the likelihood of the item being sent back.

3. Data Pipeline Integration

The scored dataset, now containing the AI-generated risk probabilities, was exported as a cleaned CSV file (`ecommerce_with_risk_scores.csv`) and seamlessly imported into Power BI Desktop via Power Query. Data types were validated, ensuring decimal structures for risk scores and prices.

4. Business Intelligence Dashboard Development

A professional, dark-themed operational dashboard was constructed in Power BI:

- **Custom DAX Measures:** Engineered calculated fields for *Total Orders*, *Total Returns*, *Return Rate %*, and *Avg Risk Score*.
- **KPI Ribbon:** Four primary card visuals positioned at the top for immediate executive visibility.
- **Categorical Analysis:** A clustered bar chart mapping Return Rate % by category to identify underperforming departments.

- **AI Risk Tracker:** A functional table matrix dynamically sorted by the Machine Learning Risk Score in descending order, enabling staff to intercept high-probability returns before shipping.

Conclusion

The successful execution of Project 5 demonstrates a robust capability to not only manipulate and analyze massive datasets but also to deploy predictive modeling in a highly practical, business-centric format. By translating raw data into a Logistic Regression model and subsequently surfacing those predictions in a highly intuitive Power BI dashboard, the final product delivers immediate operational value. It transitions the business strategy from reactive (processing returns after the fact) to proactive (identifying and managing high-risk orders in real-time).